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Core Language Working Group "tentatively ready" Issues for the February, 2021 meeting

References in this document reflect the section and paragraph numbering of document [WG21 N4868](#).

2470. Multiple array objects providing storage for one object

Section: 6.7.2 [intro.object] **Status:** tentatively ready **Submitter:** Andrey Erokhin **Date:** 2021-01-29

According to 6.7.2 [intro.object] paragraph 3,

If a complete object is created (7.6.2.8 [expr.new]) in storage associated with another object *e* of type “array of *N* unsigned char” or of type “array of *N* std::byte” (17.2.1 [cstddef.syn]), that array *provides storage* for the created object if:

- the lifetime of *e* has begun and not ended, and
- the storage for the new object fits entirely within *e*, and
- there is no smaller array object that satisfies these constraints.

The intent of the third bullet is to select a unique array object among those satisfying the first two bullets. However, it is possible to have multiple array objects of the same size satisfying the first two bullets. For example:

```
unsigned char buffer[8];
struct OhNo { std::byte data[8]; };
static_assert(sizeof(OhNo) == 8 && sizeof(int) == 4);
OhNo *p = new (buffer) OhNo;    // buffer provides storage for OhNo
int *q = new (p->data) int;     // who provides storage for this?
int *r = new (buffer + 4) int;  // who provides storage for this?
```

Proposed resolution (February, 2021):

Change 6.7.2 [intro.object] bullet 3.3 as follows:

- there is no **smaller** array object that satisfies these constraints **nested within *e***.